Fiscal Year:	FY 2022	**Task Last Updated:**	FY 02/02/2023
PI Name:	Douglas, Grace Ph.D.		
Project Title:	The Integrated Impact of Diet on Human Immune Response, the Gut Microbiota, and Nutritional Status During Adaptation to Spaceflight		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	09/01/2016	**End Date:**	03/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	6
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 3/31/2024 per F. Hernandez/ARC (Ed., 1/3/23) NOTE: End date changed to 3/31/2023 per F. Hernandez/ARC (Ed., 6/4/19) NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/19/17) NOTE: Period of performance changed to 9/1/2016-9/1/2019 (from 3/1/2015-2/28/2018) per PI, due to delayed facility access (Ed., 12/4/15)		
Key Personnel Changes/Previous PI:	July 2019 report: Collaborator changed from Vickie Kloeris (retired) to Ryan Dowdy, Ph.D., who will coordinate food upmass to ISS to support this experiment. July 2017 report: Millennia Young, Ph.D. replaced in January 2017 Robert Ploutz-Snyder, Ph.D., Pstat® as the biostatistician on this project.		

COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Lorenzi, Hernan Ph.D. (J. Craig Venter Institute - grant 80NSSC19K0930) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (University of Texas Medical Branch (UTMB)) Young, Millennia Ph.D. (NASA Johnson Space Center) Gonzalez-Juarbe, Norberto Ph.D. (J. Craig Venter Institute - grant 80NSSC19K0930)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE this project is jointly funded by NASA's Space Biology and Human Research Program (HRP) for collaborative purposes. HRP/HHC (Human Health Countermeasures element) is the originating organization for the preliminary ground study. Space Biology is the originating organization for a flight study that will follow this work. ED. NOTE (12/4/15): Period of performance changed to 9/1/2016-9/1/2019 (from 3/1/2015-2/28/2018) per PI, due to delayed facility access. Spaceflight has demonstrated concerning impacts to human physiology, including well documented immune system dysregulation. Diet, immune function, and the microbiome are interlinked, but diet is the only one of these factors that we have the ability to easily, and significantly, alter on Earth or during flight. It is expected that as we understand dietary impacts more thoroughly, we may continue to improve the spaceflight diet in direct support of crew health. We propose here an interdisciplinary study that will determine the effect of the current spaceflight diet compared to an "enhanced" spaceflight diet (25% more foods rich in omega-3 fatty acids, lycopene, flavonoids, fruits, and vegetables), with complete dietary tracking, on immune function, gut microbiota composition, and nutritional status of crew in ground-control analogs, and on International Space Station (ISS) missions. We hypothesize that by increasing consumption of fruits and vegetables and bioactive compounds (e.g., omega-3 fatty acids, lycopene, flavonoids) and enhancing overall nutritional intake during spaceflight (or a ground-based analog of spaceflight) we will improve human immunological profiles, the taxonomic profile of the gut microbiota, and nutritional status biomarkers. Positive associations between beneficial improvements will provide evidence to better predict the impact of diet on crew health and adaptation to spaceflight and to design targeted more-efficient dietary interventions. Specifically, we propose to 1) implement a pilot ground-control study with analog crew randomized between two spaceflight diets, and determine diet effects on immune dysregulation, including leukocyte distribution, inflammatory cytokine profiles, T cell function, and other relevant immunological markers, the taxonomic and metatranscriptomic profile of the gut microbiome, and nutritional biomarkers and metabolites at selected intervals, and statistically evaluate to associate shifts across each measure and in relation to nutrient intake; and 2) determine effects of diet on immune dysregulation, including leukocyte distribution, inflammatory cytokine profiles, T cell function, and other relevant immunological markers, the taxonomic and metatranscriptomic profile of the gut microbiome, and nutritional biomarkers and metabolites at selected intervals in spaceflight and statistically evaluate to associate shifts across each measure, and in relation to ground-control data. This study specifically addresses the B.4 Research Emphasis 3 - Understanding How Complex Organisms Adapt to Spaceflight, by providing detailed comprehensive analysis of the effect of diet on the human immune system, gut microbiota, and nutritional status during human adaptation to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improvements in immune and nutritional status and the associated microbiome will provide evidence of the impact of diet on crew health and adaptation to this spaceflight analog, and in spaceflight. Beneficial associations between biomarkers and the diet will aid in the design and development of more efficient targeted dietary interventions for exploration missions. The proposed investigation will provide one of the most comprehensive and interdisciplinary human diet, immune, and gut microbiome studies completed in microgravity or on Earth. The outcome will have significant scientific and medical applications for space travelers, and for Earth-bound populations.
Task Progress:	In the past year we have had 3 subjects (including a 1-year subject) complete all baseline data collections (BDC) and in-flight Food Physiology biological and data collections. To date, 6/9 subjects completed all data collections. We have 1 subject on the International Space Station (ISS) currently. We completed BDCs for the 2 final subjects who will be on ISS starting in spring 2023. The NASA Human Exploration Research Analog (HERA) Food Physiology result have been published in Scientific Reports (Ed. Note: See Bibliography). In just 45 days, results show a statistically significant benefit in several health-related measures and cognitive outcomes for subjects consuming the enhanced diet compared to the standard diet. This has important implications, especially as programs request food system reductions (mass/volume/preparation equipment) even from the standard diet. The data indicates the need to study the impacts of food system changes before flight implementation.
Bibliography Type:	Description: (Last Updated: 11/22/2025)
Articles in Peer-reviewed Journals	Douglas GL, DeKerlegand D, Dlouhy H, Dumont-Leblond N, Fields E, Heer M, Krieger S, Mehta S, Rooney BV, Torralba MG, Whiting SE, Crucian B, Lorenzi H, Smith SM, Young M, Zwart SR. "Impact of diet on human nutrition, immune response, gut microbiome, and cognition in an isolated and confined mission environment." Sci Rep. 2022 Dec 15;12:20847. https://doi.org/10.1038/s41598-022-21927-5 ; PMID: 36522361; PMCID: PMC9755260 , Dec-2022
Articles in Peer-reviewed Journals	Douglas GL, Cooper MR, Wu H, Gaza R, Guida P, Young M. "Impact of galactic cosmic ray simulation on nutritional content of foods." Life Sci Space Res. 2021 Feb;28:22-5. http://www.sciencedirect.com/science/article/pii/S2214552420300869 ; PMID: 33612176 , Feb-2021